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CamLoop4D: Controllable 4D Scene Generation by Closing the Camera Loop between Generation and Reconstruction

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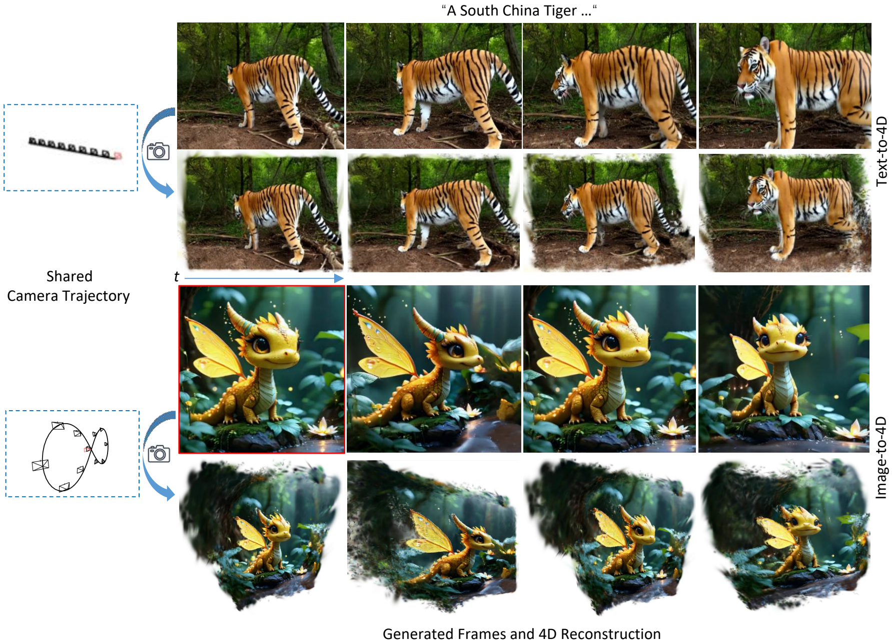

Figure 1: *CamLoop4D generates controllable 4D scenes from a text prompt or a single image under user-specified camera trajectories; the red border indicates the input conditioning image for the Image-to-4D task. By sharing the same Plücker trajectory tensor between a camera-conditioned video diffusion model and a 4D Gaussian reconstructor, our co-design framework keeps the two stages geometrically aligned without pose re-estimation, producing temporally coherent 4D content on a single RTX 3090 in ∼40 minutes.*

Abstract

Generating dynamic 3D scenes with coherent geometry and motion from a text prompt or a single image, under user-specified camera trajectories, is a long-standing goal in computer graphics content creation. Existing pipelines cascade a video generator and a 4D reconstructor through an error-prone pose-estimation step, where estimation errors compound with generation artifacts. We present CamLoop4D, a generation-reconstruction co-design *framework that closes this loop: a camera-conditioned video diffusion model and a 4D Gaussian reconstructor operate on the* same *Plücker trajectory tensor, removing pose re-estimation by construction, and the interface is generator-agnostic. For reconstruction, we introduce* hybrid motion bases *that combine six fixed $\mathfrak{se}(3)$ generators encoding rigid-body priors with learnable non-rigid bases, reducing motion-bank storage from $\mathcal{O}(B \cdot F)$ to $\mathcal{O}(B)$ while improving fidelity; under the matched-input protocol (all methods given the same real video and shared poses), the reconstruction module surpasses the strongest monocular baselines. CamLoop4D achieves the best PSNR/SSIM among all compared methods on the iPhone benchmark and the highest user-preference rates against seven recent baselines in text-to-4D and image-to-4D evaluation, and it generates an 80-frame 960×720 scene in ∼40 minutes on a single RTX 3090 with real-time 20 fps playback, bringing controllable 4D content creation within reach of individual creators. Our implementation is available at https://github.com/lihao1477024473/CamLoop4D.*

CCS Concepts

• ***Computing methodologies*** → ***Animation;*** *Rendering; Shape modeling;*

Keywords: *4D scene generation; Camera-controlled video diffusion; 3D Gaussian splatting; Spatial intelligence*

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1. Introduction

Dynamic 3D content lies at the heart of modern computer graphics, powering virtual production, games, visual effects, and immersive VR/AR experiences, yet authoring it remains labour-intensive: artists must model geometry, rig motion, and choreograph cameras. A long-standing goal is therefore to generate dynamic 3D scenes (i.e., 4D content) with full spatiotemporal consistency from minimal input, such as a text prompt or a single image, while retaining the camera-level control that graphics workflows demand.

Despite significant advances in static 3D reconstruction [MST*21, KKLD23], synthesizing dynamic 3D scenes with semantic and temporal coherence from a single image or text, guided by user-specified camera trajectories, remains open. Existing dynamic reconstruction methods typically require synchronized multi-view videos [FKMW*23, LKLR24, WYF*24] or depth sensors [GWW*19, PBM20, WLPL22, WLHJ*20], and scene-level navigation approaches [YDH*24, YDH*25] offer only step-wise control without fine-grained trajectory conditioning.

Recent camera-controlled video diffusion models [HXG*25, ZLJ*24, HC25, BSS*25, BSQ*25, ZGV*25] and 4D reconstruction via 3D Gaussian splatting [KKLD23] supply the necessary building blocks, but *naively cascading* the two stages creates a critical bottleneck: an intermediate pose-estimation step introduces errors that compound with generation artifacts. Moreover, existing dynamic representations are either fully unconstrained and prone to overfitting, or overly rigid and unable to capture complex deformations.

We observe that the root cause of this bottleneck is a *design mismatch*: the camera parameters that already condition the generative model are discarded and re-estimated before reconstruction. CamLoop4D eliminates this mismatch through **generation-reconstruction co-design**, in which both stages consume the same Plücker trajectory tensor, so the reconstructor reads the specified camera parameters directly instead of recovering them post hoc. We are precise about scope. Sharing the trajectory removes the pose-*estimation stage* from the pipeline, though it does not, by itself, force the generated frames to obey the specified path perfectly (we quantify the residual generator deviation in Section 3.2.3). To our knowledge no prior open-prompt 4D system reuses the conditioning trajectory in this way. Our contribution is to formalise this interface, quantify its benefit, and validate it across generators. Both stages remain frozen and modular, because the co-design is an interface contract rather than joint training — precisely what keeps the pipeline practical on consumer hardware and portable to future generators. Our contributions are:

(1) **A shared-Plücker-trajectory interface** (a one-way convention, not joint training or feedback) treating cross-stage trajectory consistency as an explicit design axis. It yields +0.88 dB PSNR over poses from the *static* estimator DUSt3R [WLC*24], or a still-positive +0.34 dB against dynamic-aware MonST3R; the generator is used *frozen, off-the-shelf*, so the interface, not a new generator, is the contribution [BSQ*25, HXG*25].

(2) **Hybrid motion bases for persistent 3D Gaussians** that combine six fixed $\mathfrak{se}(3)$ generators encoding rigid-body priors with learnable bases. With per-Gaussian per-frame coefficients the six generators already span the instantaneous motion space; the trainable bases act as a shared, scene-adapted low-rank prior that *conditions and regularises* the optimisation. This raises reconstruction fidelity by +0.64 dB PSNR over a fully-learnable basis set, and under matched real-video inputs improves on SoM by +0.52 dB and on the strongest baseline MoSca by a small but significant aggregate margin (+0.19 dB PSNR), at the cost of a larger per-frame coefficient tensor: a deliberate expressiveness-for-memory trade we quantify in Section 3.2.2. The advantage replicates on a second dynamic benchmark and is independent of the generative front-end.

(3) **A practical single-GPU system** for text/image-to-4D generation featuring unified foreground-background modelling and Gaussian downsampling. CamLoop4D completes an 80-frame 960×720 sequence in $\sim$40 minutes on a single RTX 3090 while rendering at 20 fps, over an order of magnitude cheaper than methods requiring multi-GPU clusters [WGP*25], bringing high-quality 4D content creation within reach of individual artists and small studios.

2. Related Work

Video Diffusion Models and Camera Control. Latent diffusion models [RBL*22, SME22] have been extended from images to video [HSG*22, HCS*22, HYZ*22, HDZ*22, CXH*23, ZWZ*23, BPH*24], with adapter-based approaches such as AnimateDiff [GYR*24b] enabling motion priors to be grafted onto pre-trained image generators. Building on these backbones, a rapidly growing line of work injects explicit camera control: CameraCtrl [HXG*25], CamI2V [ZLJ*24], VD3D [BSS*25], AC3D [BSQ*25], and Stable Virtual Camera [ZGV*25] encode camera trajectories as Plücker embeddings and inject the extracted multi-scale features into the (temporal) attention layers of video diffusion models, while training-free alternatives manipulate latents directly [HC25]. These models enable controllable camera motion for filmmaking and VR/AR applications [CWX*23, GYR*24a, Hu24, KMT*23, XZL*24]. However, they operate purely in pixel space and lack an explicit 3D representation, so their outputs cannot be re-rendered from novel viewpoints, edited at the object level, or played back in real time. Crucially, when such videos are fed to a downstream reconstructor, the camera parameters that conditioned the generation are typically discarded and re-estimated, a redundancy that CamLoop4D eliminates.

Dynamic Scene Reconstruction. Classical non-rigid reconstruction relies on RGB-D sensors [NFS15, DKD*16, IZN*16, ZNI*14, BZTN20] or hand-designed priors [KDL17, RVCK16, RYA14], while multi-view rigs enable high-quality volumetric video [BFO*20, LSZ*22, IRG*23, WCS*22]. Neural radiance fields [MST*21] spurred a family of dynamic extensions, including deformation-field methods [PSB*21, PSH*21, PCPMMN21], space-time radiance fields [XHKK21, LNSW21, WELG21, FKMW*23, CJ23, SCL*23], and image-based rendering for long videos [LWC*23]; monocular settings additionally exploit consistent video depth [YKG*20, LHS*20, KRH21, ZCT*21]. 3D Gaussian Splatting [KKLD23] brought real-time rendering to this problem, with dynamic variants based on per-frame Gaussians [LKLR24], deformation networks [WYF*24, YGZ*24], 4D primitives [YYPZ24, DWD*24], and compact motion bases [KLD24, LCLX24]. Most recently, SoM [WYG*25] and MoSca [LWH*24] substantially improved monocular reconstruction by incorporating auxiliary signals from large vision models, e.g., segmentation [KMR*23], depth [YKH*24, PYS*24], and point tracking [DYV*23, KRG*24]. While these methods perform well on captured videos, they cannot generate 4D content from text or images, and they invariably depend on pose estimation [TD21, WLC*24] whose errors propagate into the reconstruction. CamLoop4D bridges this gap by sharing the Plücker trajectory tensor across generation and reconstruction, removing the pose re-estimation *stage* from the pipeline (without claiming the generated frames follow the path perfectly; cf. Section 3.2.3); its hybrid motion bases further advance reconstruction quality itself (+0.52 dB over SoM under matched inputs).

3D and 4D Generation. Early 3D generation targeted static objects [ADMG18, PFS*19, GSW*22] or static scenes [CWL23, LLM*23], with score distillation [PJBM23] enabling text-driven synthesis. Subsequent work added temporal deformation for object-level 4D generation [SSP*23, ZLN*24, RPT*23, ZYX*23, YXW*23, RXM*24, SGW*24, PYZZ26], which, however, produces isolated objects without scene context or camera-path control. Scene-level navigation approaches [YDH*24, YDH*25] offer step-wise exploration but lack fine-grained trajectory conditioning. Recent scene-level 4D generators include DimensionX [SCL*25], GenXD [ZLL*25], 4Real [YWZ*24], CAT4D [WGP*25], and Free4D [LHC*25]. CAT4D achieves high quality but requires 8×A100 GPUs, whereas Free4D is tuning-free but re-estimates camera poses instead of sharing them with the reconstruction stage. In contrast, CamLoop4D identifies *cross-stage trajectory consistency* as the key missing axis. A frozen video model is coupled with the reconstructor through the same Plücker trajectory, closing the loop that cascaded pipelines leave open, and the purely Plücker-level interface is validated with two generators and extensible to future camera-conditioned models.

3. Method

We target 4D scene generation from open natural language prompts or single images. Figure 2 illustrates the overall pipeline of CamLoop4D, comprising two tightly coupled stages: camera-controlled video generation (Section 3.1) and 4D scene reconstruction (Section 3.2). The central design principle, *generation-reconstruction co-design*, manifests in three ways: (i) both stages consume the same Plücker trajectory tensor, removing the pose re-estimation *stage* from the pipeline (we quantify the residual generator deviation in Section 3.2.3); (ii) the reconstruction stage introduces hybrid motion bases that combine rigid-body priors with learnable deformation, providing a structured yet expressive 4D representation; and (iii) the interface is generator-agnostic, depending only on the Plücker tensor and not on any specific generator architecture. This eliminates the fragile pose-estimation bridge of cascaded pipelines while running entirely on a single consumer GPU.

3.1. Camera-Controlled Video Generation

Camera Representation. Following CameraCtrl [HXG*25], we adopt Plücker embeddings [SRF*21] to encode camera trajectories. For each pixel (u,v), the Plücker embedding is:

$$p_{u,v} = \left[\, o \times d_{u,v} \,;\, d_{u,v} \,\right] \in \mathbb{R}^6, \tag{1}$$

where $[\,\cdot\,;\,\cdot\,]$ denotes *concatenation* of the two 3-vectors (not an inner product), $o \in \mathbb{R}^3$ is the camera center in world coordinates, and $d_{u,v}$ is the normalized ray direction computed from the camera intrinsics $\mathbf{K}$ and extrinsics $[\mathbf{R}|\mathbf{t}]$:

$$d_{u,v} = \text{normalize}\left(\mathbf{R}\mathbf{K}^{-1}[u,v,1]^T\right), \tag{2}$$

with $\mathbf{R} \in \mathbb{SO}(3)$ and $\mathbf{t} \in \mathbb{R}^3$. The full camera trajectory is then represented as a sequence $\{P_i\}_{i=1}^n$ with $P_i \in \mathbb{R}^{6\times h\times w}$.

Video Generation. We leverage the pre-trained camera-controlled video diffusion model AC3D [BSQ*25], which builds upon CogVideoX [YTZ*25]. AC3D uses a learned encoder to extract multi-scale features from the Plücker embeddings and injects them into the temporal attention layers of the diffusion model. The standard latent diffusion objective is:

$$L(\theta) \triangleq \mathbb{E}_{z,c,\varepsilon\sim\mathcal{N}(0,I),t}\left[\|\varepsilon - \varepsilon_\theta(z_t,c,t)\|_2^2\right], \tag{3}$$

where $z = \mathcal{E}(x)$ is the latent video encoding, c is the conditioning input (text prompt or reference image), ε_θ is the denoiser, and $z_t = \alpha_t z_0 + \sigma_t\varepsilon$ with $\sigma_t = \sqrt{1-\alpha_t^2}$ [HJA20].

We use the publicly available AC3D checkpoint without additional fine-tuning; the trajectory tensor $\{P_i\}$ produced here is *directly shared* with the reconstruction stage (Section 3.2.3).

3.2. 4D Scene Reconstruction

3.2.1. Preliminaries

We represent dynamic scenes using persistent 3D Gaussians [KKLD23, WYG*25]. Each Gaussian in the canonical frame t_0 is parameterised by $g_0 = (\mu_0, R_0, s, o, c)$, where $\mu_0 \in \mathbb{R}^3$ is the mean, $R_0 \in \mathbb{SO}(3)$ the rotation, $s \in \mathbb{R}^3$ the scale, o the opacity, and $c \in \mathbb{R}^3$ the colour (s, o, c are time-invariant). Rendering follows the standard splatting pipeline [KKLD23]: each Gaussian is projected onto the image plane via an affine approximation and alpha-blended to produce RGB images and depth maps.

3.2.2. Dynamic Scene Representation

For dynamic scene representation and motion modelling, inspired by SoM [WYG*25], we design a persistent 3D Gaussian representation. Motion is modelled using a set of compact and globally shared hybrid motion bases (each a $\mathfrak{se}(3)$ Lie-algebra element, converted to $\mathbb{SE}(3)$ by the exponential map), with each Gaussian trajectory expressed as a linear combination thereof, enabling efficient representation of complex dynamic behaviours.

To model dynamic 3D scenes, we maintain a set of N canonical 3D Gaussian distributions, whose positions and orientations are transformed over time via rigid transformations to simulate motion. The transformation from the canonical frame t_0 to t is denoted by $T_{0:t} = [R_{0:t}; t_{0:t}] \in \mathbb{SE}(3)$. The pose of each Gaussian at t, given by (μ_t, R_t) is computed as:

$$\begin{aligned} \mu_t &= R_{0:t}\mu_0 + t_{0:t}, \\ R_t &= R_{0:t}R_0, \end{aligned} \tag{4}$$

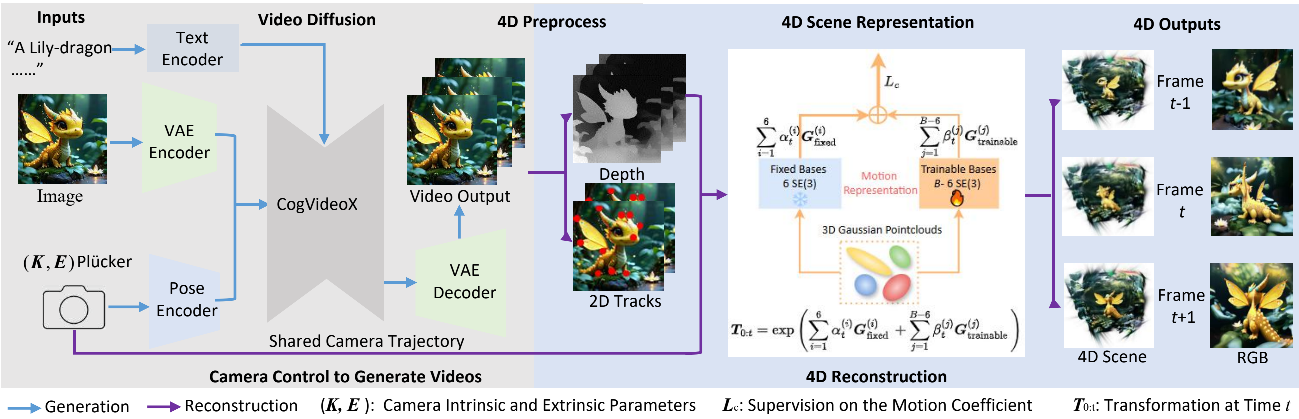

Figure 2: *System Overview. We first encode a text prompt or single image into a latent spatial representation using a VAE encoder. Concurrently, based on camera intrinsic* **K** *and extrinsic* **E** *parameters, the specified trajectory is encoded using Plücker coordinates and shared by the generation and reconstruction stages, removing pose re-estimation. AC3D [BSQ*25] (Section 3.1) then generates a video sequence with the desired camera motion; off-the-shelf models [DYV*23, YKH*24] extract depth maps and 2D point trajectories, which, with the RGB frames, feed reconstruction. Dynamic scenes are represented by persistent 3D Gaussians whose motion is a linear combination of globally shared, compact hybrid motion bases (each an $\mathfrak{se}(3)$ element mapped to $\mathbb{SE}(3)$ via the exponential map) for efficient modelling of complex dynamics (Section 3.2).*

where μ_t and R_t represent the 3D position and rotation of the Gaussian at t.

At any time t, the transformation $T_{0:t} \in \mathbb{SE}(3)$ is computed via the exponential map of a weighted sum of $\mathfrak{se}(3)$ basis elements:

$$T_{0:t} = \exp\left(\sum_{i=1}^{6} \alpha_t^{(i)} G_{\text{fixed}}^{(i)} + \sum_{j=1}^{B-6} \beta_t^{(j)} G_{\text{trainable}}^{(j)}\right), \tag{5}$$

where $G_{\text{fixed}}^{(i)} \in \mathfrak{se}(3)$ are the six frozen generators (three translations and three rotations, see Appendix) and $G_{\text{trainable}}^{(j)} \in \mathfrak{se}(3)$ are learnable bases. The coefficients $\alpha_t^{(i)}$ and $\beta_t^{(j)}$ are per-Gaussian and per-frame; we omit the Gaussian index n in Equation (5) for readability (it is made explicit in Equation (6)). The set of motion bases is globally shared across all Gaussians [KLD24]; only the coefficients are per-Gaussian.

Mathematical justification. Since $\mathfrak{se}(3)$ is a vector space, any linear combination $\xi = \sum_b \gamma^{(b)} G^{(b)}$ remains in $\mathfrak{se}(3)$, and $\exp(\xi) \in \mathbb{SE}(3)$ is a valid rigid transformation. Although $\mathbb{SE}(3)$ is non-commutative, the motion coefficient loss (Equation (6)) keeps coefficients small (mean absolute magnitude 0.037 ± 0.021, 95th percentile 0.082 across all 14 iPhone sequences), ensuring the first-order BCH approximation holds reliably. Each basis is compactly stored as a 6D vector and converted to $\mathbb{SE}(3)$ via the exponential map during the forward pass [WYG*25].

Role of the trainable bases. Two notions are easily conflated. (i) *Expressiveness*: with per-Gaussian *per-frame* coefficients the six fixed generators (which already span $\mathfrak{se}(3)$) already suffice to give every Gaussian an arbitrary rigid transform at every frame, so the trainable bases add no instantaneous degrees of freedom; scene non-rigidity comes from different Gaussians moving differently. (ii) *Conditioning*: the trainable bases act as a shared, scene-adapted low-rank dictionary that couples Gaussians, letting coherent motion be explained by a few globally shared directions rather than fitting noisy per-Gaussian coefficients independently against imperfect supervision. This is the shared-low-rank-motion prior of compact motion-basis methods [KLD24, LCLX24, WYG*25]; our addition is the fixed rigid scaffold, justified empirically (Section 4.3). Removing the trainable bases lets the per-frame coefficients overfit noisy tracks and collapses PSNR by $\sim$5 dB; this is a conditioning failure, not lost expressiveness (multi-seed controls, an alternative-regulariser comparison, a temporal hold-out, and the zero-init/symmetry-breaking details are in the supplement).

Comparison with SoM and storage. SoM [WYG*25] stores time-varying bases $B \in \mathbb{R}^{B\times F\times 6}$ with time-invariant coefficients $C \in \mathbb{R}^{N\times B}$; we move the time dependence into the coefficients, storing static bases $B \in \mathbb{R}^{B\times 6}$ with per-Gaussian per-frame coefficients $C \in \mathbb{R}^{N\times F\times B}$. The basis bank thus shrinks from $\mathcal{O}(B\cdot F)$ to $\mathcal{O}(B)$, but this is *not* a net storage win: the $\mathcal{O}(N\cdot F\cdot B)$ coefficient tensor now dominates ($N \gg F, B$), making our total motion state $\approx$240 MB versus SoM's $\approx$3 MB (supplement). We make no compactness claim; we trade $\approx$80$\times$ larger (but still small, well under the 24 GB budget) motion state for the per-frame expressiveness that, with the rigid scaffold, yields the +0.52 dB gain over SoM under matched inputs (Table 2).

3.2.3. Shared Camera Trajectory Interface

We share the Plücker trajectory $\{P_i\}$ from Section 3.1 directly with the reconstructor, which therefore consumes the specified poses rather than re-estimated ones. This removes the estimation step whose errors dominate cascaded pipelines (it does not force the generated frames to obey the path). Both stages share one convention: the intrinsics **K** that condition the generator, a world frame at the canonical frame t_0, and the metric scale fixed by the specified trajectory; Depth Anything [YKH*24] depth is least-squares shift/scale-aligned per frame and TAPIR [DYV*23] tracks are lifted with the same intrinsics, so no separate scale or frame resolution is needed. TAPIR queries are sampled uniformly over the frame with no foreground restriction, 256 per sequence, and back-projected with the shared intrinsics and aligned depth. The shared low-rank motion prior (Section 3.2.2) absorbs noisy or sparse track supervision, so reconstruction is not sensitive to the exact track density (removing the trainable bases collapses held-out PSNR by ∼5 dB, Section 4.3; supervision-robustness regimes are analysed in Supp. C.7). As a diagnostic only (computed offline with COLMAP, never fed back), the residual AC3D deviation is small (ATE 0.021 m, angular 1.3°); being COLMAP-based on dynamic content, we read it as an indicative upper bound, not a precise measurement. Sharing yields +0.88 dB over DUSt3R [WLC*24]-estimated poses (Table 6). Since DUSt3R is a weak static estimator (0.142 m ATE), we also re-estimate from the same frames with stronger ones: the gain shrinks to +0.53 dB (masked COLMAP) and +0.34 dB (dynamic-aware MonST3R [ZHT*24]) but persists, and sharing additionally avoids the estimator's runtime and failure modes (full table in the supplement). To quantify robustness to generator-induced trajectory inconsistency, we perturb the poses fed to the reconstructor by a controlled ATE (supplement): the loss is <0.05 dB at AC3D's residual (≈0.02 m), reaches only ≈ 0.8 dB at 0.10 m, and decays sharply only beyond ≈0.2 m — well past AC3D's residual and beyond where estimation error would sit (DUSt3R 0.142 m). Residual inconsistencies therefore surface as LPIPS texture smoothing rather than geometric drift. The interface is generator-agnostic: replacing AC3D with CameraCtrl [HXG*25] costs only −0.47 dB (supplement).

3.2.4. Unified Modelling and Downsampling

Unlike SoM [WYG*25], which separates foreground and background, we treat both as co-moving entities under a unified dynamic representation with shared motion bases, better suited for camera-controlled generation where everything moves; instance segmentation [KMR*23] delineates regions within the same basis framework. To prevent excessive Gaussian density near scene boundaries, we downsample dynamic Gaussians by 0.5× during training, improving boundary coherence and reducing memory usage.

3.3. Optimization

Training Details. We use Adam with learning rate 10^{-4}, 1,000 iterations of initial fitting followed by 600 epochs of joint optimisation. We set $B = 15$ motion bases (6 fixed + 9 trainable), selected via ablation (Section 4.3), initialise 50,000 Gaussians with adaptive density control [KKLD23], and apply 0.5× downsampling. All experiments run on a single RTX 3090.

Computational Efficiency. CamLoop4D's end-to-end time on a single RTX 3090 breaks down as: video generation (∼5 min), preprocessing (∼5 min), and 4D reconstruction (∼30 min), totaling ∼40 min at 20 fps rendering, with a peak GPU memory of ∼11 GB during optimisation and ∼3 GB at render time (the ≈240 MB motion state is a small part of this). For a hardware-normalised view, CamLoop4D costs ≈0.7 GPU-hours (RTX 3090); CAT4D [WGP*25] reports 8×A100 for ∼1 h (≈8 A100-GPU-hours), Free4D [LHC*25] 1×A100 for ∼2 h, and SoM [WYG*25] 1×A6000 for ∼1 h. We report these as the methods' stated operating points, not strict same-hardware benchmarks; even discounting the GPU-class difference, the order-of-magnitude gap to multi-GPU systems is clear.

We introduce additional supervision on the motion coefficients via a motion coefficient loss applied to each Gaussian n at each frame f:

$$L_c := \frac{1}{NF}\sum_{n=1}^{N}\sum_{f=1}^{F}\left[(1-\lambda)\sum_{i=1}^{6}\left(\alpha_{n,f}^{(i)}\right)^2+\lambda\sum_{j=1}^{B-6}\left(\beta_{n,f}^{(j)}\right)^2\right], \quad (6)$$

where $\alpha_{n,f}^{(i)}$ and $\beta_{n,f}^{(j)}$ are the fixed-basis and trainable-basis coefficients of the n-th Gaussian at frame f, N is the number of Gaussians, F the number of frames, and $\lambda = 0.8$. The overall training loss is $L = L_{\text{rgb}} + \mu_d L_{\text{depth}} + \mu_t L_{\text{track}} + \mu_c L_c$, where L_{rgb}, L_{depth}, and L_{track} are the photometric, depth, and trajectory supervision terms following SoM [WYG*25], with $\mu_d = 0.5$, $\mu_t = 0.2$, and $\mu_c = 0.01$. The asymmetric weighting (λ=0.8) more strongly damps the learnable coefficients, encouraging the optimiser to explain motion with rigid generators first.

4. Experiments

We evaluate CamLoop4D on 4D reconstruction (iPhone/DyCheck and NVIDIA Dynamic Scenes), text/image-to-4D generation against seven recent baselines, cross-generator portability, and content-creation applications. Static figures cannot convey temporal coherence; please see the *supplementary video*.

4.1. 4D Reconstruction

Protocol. We use the iPhone dataset [GLT*22] (14 sequences) with PSNR, SSIM [WBSS04], and LPIPS [GLT*22, HZZ*20]. Following the DyCheck protocol these are *held-out novel-view* metrics (the model is fit on the training camera and scored on held-out co-visible cameras), so they test 3D structure, not training-view fit. We report unmasked full-frame metrics over all 14 sequences, so our values are systematically lower than published DyCheck mPSNR figures (≈19.3 with provided poses, and ≈18.8–19.0 even with no pose input, for MoSca), which are computed under the co-visibility mask on a 7-sequence multi-camera split; the two are therefore not directly comparable. The *matched-input* comparison is our controlled experiment and the sole basis for architectural claims: all methods receive the same real video with the same DUSt3R [WLC*24] poses. The *system-level* comparison instead places each pipeline in its deployment setting (generated video for us, real for baselines); since inputs differ it is deployment context, not a controlled comparison. The system-level protocol is detailed in Supp. C.4: AC3D is conditioned on only the *real first frame* and the sequence's *ground-truth trajectory* at the same frame count and resolution, so the generated and real frames correspond one-to-one and are scored against the real held-out frames *without* any search, warping, or colour matching. The scenes are *not* treated as static, and the metrics are *not* computed on training views.

Matched-input (primary). Our module reaches 16.41 dB / 0.621 / 0.440 (Table 2), +0.52 dB over SoM and +0.18 dB over MoSca. We are candid that the MoSca margin is small and *non-uniform*: per scene (full table in the supplement) CamLoop4D wins 11 of 14 but

Table 1: *Matched-input reconstruction on a* second *benchmark, NVIDIA Dynamic Scenes [YKG*20] (7 monocular sequences), same protocol as the iPhone matched-input experiment. The ordering replicates. Best in* ***bold****.*

Method	PSNR↑	SSIM↑	LPIPS↓
SoM [WYG*25]	21.24	0.71	0.21
MoSca [LWH*24]	21.74	0.73	0.19
Ours (recon. only)	**21.94**	**0.74**	**0.18**

loses the three most-articulated scenes (*block*, *hand*, *mochi-high-five*), where MoSca's separate foreground/background helps. The per-sequence paired test gives +0.19 dB over MoSca ($t(13) = 2.72$, $p = 0.009$, one-sided) and +0.52 dB over SoM ($t(13) = 6.11$). Our claim, scoped in Section 5, is a small but significant aggregate gain from the hybrid bases and unified modelling, independent of the generator. Raw per-run logs and the eval script will be released.

Second benchmark. To show the advantage is not specific to iPhone statistics, we repeat the matched-input protocol (same real video and DUSt3R poses for all methods) on the NVIDIA Dynamic Scenes dataset [YKG*20] (7 monocular sequences, Table 1). The ordering replicates: +0.20 dB over MoSca and +0.70 over SoM, with the MoSca margin at the edge of significance at this small sample size ($t(6) = 1.87$, $p = 0.055$). We read this as corroborating the *direction* of the effect on a second dataset, not as a second strong-significance claim.

System-level (deployment). The system-level comparison is deployment context, not a controlled comparison (protocol in Supp. C.4). From *generated* video CamLoop4D reaches 16.55 dB / 0.62 (Table 2), on par with real-video baselines (MoSca 16.23, SoM 15.89). This is not like-for-like; the rough parity arises because sharing poses avoids the ≈0.88 dB DUSt3R penalty (Table 6) that offsets the lower fidelity of generated frames. The higher LPIPS (0.48 vs. 0.45) is generated-video texture smoothing, not architecture (our matched-input LPIPS is at parity with MoSca, 0.440 vs. 0.443). The interface is also generator-agnostic: swapping AC3D for CameraCtrl [HXG*25] costs only −0.47 dB (supplement).

Qualitative. In Figure 3, 4D Gaussians and HyperNeRF are blurry with geometric distortion and MoSca/SoM retain view-dependent artifacts, whereas CamLoop4D recovers cleaner geometry and more stable high-frequency detail (e.g. the flower-pot pattern, the rolling arm). It also preserves dynamic layout and object identity under viewpoints *outside* the input trajectory (qualitative novel-view renderings in the supplement; no ground truth exists for extrapolated views in the monocular setting, so the quantitative 3D support is the held-out DyCheck metrics above).

4.2. 4D Generation

Setup. We compare text-to-4D against Free4D [LHC*25], CAT4D [WGP*25], 4Real [YWZ*24], 4Dfy [BSR*24], and Dream-in-4D [ZLN*24], and image-to-4D against Free4D, GenXD [ZLL*25], and DimensionX [SCL*25], all on identical prompts. We report VBench [HHY*24] (a 2D proxy that only indirectly probes 3D geometry) on per-baseline pairwise prompts and an independent 25-prompt unified benchmark, plus a user study. All open-source baselines are re-run from official checkpoints; CAT4D

Table 2: *iPhone reconstruction, held-out novel view (PSNR / SSIM / LPIPS).* Top: *matched-input (primary, controlled), all methods given the same real video and DUSt3R poses.* Bottom: *system-level (deployment), inputs differ per method, so not a controlled comparison. Our row reconstructs from AC3D video generated from the real first frame and the ground-truth trajectory (Supp. C.4); scenes are not static and metrics are not on training views. Best in* ***bold****.*

Method	PSNR↑	SSIM↑	LPIPS↓
Matched-input (controlled)			
MoSca [LWH*24]	16.23	0.616	0.443
SoM [WYG*25]	15.89	0.602	0.459
Ours	**16.41**	**0.621**	**0.440**
System-level (deployment; inputs differ)			
4DGS [WYF*24]	15.42	0.58	0.45
HyperNeRF [PSH*21]	15.99	0.59	0.51
MoSca [LWH*24]	16.23	0.61	0.45
SoM [WYG*25]	15.89	0.60	0.46
Ours *(generated input)*	16.55	0.62	0.48

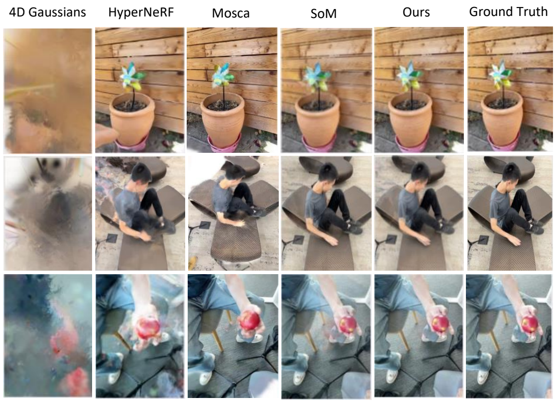

Figure 3: *Visual comparison of reconstruction quality on the iPhone dataset.*

has no public code, so its numbers are transcribed and flagged *not equal-footing*, used for no acceptance-critical claim (provenance in supplement).

User study. 38 blinded participants (hidden labels, randomised left/right) each made 15 forced-choice comparisons of same-path rendered videos across the seven baseline pairs (570 judgements, 60–82 per pair). With Holm–Bonferroni correction over the seven pairs, CamLoop4D is preferred significantly over every baseline except CAT4D (68%, uncorrected $p = 0.041$, not significant after correction); full counts, Wilson CIs, and the expert/non-expert split are in the supplement.

Results. On text-to-4D (Table 4, Figure 4, Figure 5) CamLoop4D attains the highest Aesthetic across all pairs and large Dynamic gains over Free4D (+12.3 pp) and 4Real (+14.8 pp); against the strongest baseline CAT4D it is higher Aesthetic (64.4 vs. 61.9) at >10× less compute, with CAT4D marginally ahead on Consistency/Dynamic. On image-to-4D (Table 5, Figure 6) it leads Dynamic over Free4D (+11.7) and Consistency over GenXD (+6.6) and is highest Aesthetic. The unified benchmark agrees (96.4 / 47.8

Table 3: *Geometric self-consistency of* generated *4D scenes on the 25-prompt benchmark (no real GT; lower is better). Cam-ATE = camera-following error, MV-Depth = multi-view depth disagreement, Reproj = cross-view reprojection error.*

Method	Cam-ATE (m)↓	MV-Depth (%)↓	Reproj (px)↓
Free4D (cascade)	0.112	7.9	2.9
CamLoop4D (ours)	**0.024**	**4.8**	**1.6**

Table 4: *Text-to-4D pairwise VBench [HHY*24] (%). *CAT4D not equal-footing. Dream4D =Dream-in-4D. Cons. = Consistency, Dyn. = Dynamic Degree, Aes. = Aesthetic, TA = Text Alignment.*

Method	TA	Cons.	Dyn.	Aes.
Free4D vs Ours	26.2 I 26.1	95.5 I 95.8	33.3 I 45.6	60.5 I 62.3
CAT4D* vs Ours	25.9 I 26.1	97.7 I 96.9	52.2 I 48.4	61.9 I 64.4
4Real vs Ours	26.0 I 26.1	95.7 I 96.1	33.7 I 48.5	51.9 I 64.7
4Dfy vs Ours	25.8 I 26.0	91.3 I 96.7	52.4 I 53.9	55.6 I 61.8
Dream4D vs Ours	25.3 I 25.7	91.6 I 94.4	53.4 I 53.5	56.2 I 65.4

Table 5: *Image-to-4D pairwise VBench [HHY*24] (%).*

Method	Cons.	Dyn.	Aes.
Free4D vs Ours	96.9 I 97.1	40.4 I 52.1	60.5 I 62.3
GenXD vs Ours	90.2 I 96.8	98.3 I 99.5	38.2 I 40.9
DimensionX vs Ours	97.7 I 96.8	22.2 I 22.1	55.0 I 57.6

Table 6: *Ablation on the iPhone dataset (14 seqs, mean over 3 seeds). All rows use the* same *AC3D-generated frames; "w/o Shared Camera" runs DUSt3R on those frames, so the* +0.88 *dB isolates pose source on identical input.*

Configuration	PSNR↑	SSIM↑	LPIPS↓
Full model (hybrid)	**16.55**	**0.62**	**0.48**
Fully-learnable bases (no fixed)	15.90	0.61	0.49
w/o Trainable Bases (fixed-only)	11.52	0.58	0.61
w/o Shared Camera	15.67	0.59	0.52
w/o Motion Coeff. Loss	15.93	0.61	0.49
w/o Unified Modelling	16.53	0.61	0.50
w/o Downsampling	16.48	0.58	0.51

/ 63.1 vs. CAT4D 96.1 / 49.5 / 61.2; supplement), so the trends are not prompt-dependent.

Geometric consistency of generated scenes. VBench and the same-path user study are 2D proxies, so we add a direct geometric check on the *generated* (not reconstructed-from-real) scenes, comparing against the cascade baseline Free4D [LHC*25] on the 25-prompt benchmark (Table 3). Since no real ground truth exists for generated content, we measure three *self-consistency* quantities: camera-following error (ATE between the rendered scene's realised path and the specified path), multi-view depth consistency (relative depth disagreement across overlapping views), and cross-view reprojection error. CamLoop4D is markedly more 3D-consistent on all three, with a camera-following error $\sim 5\times$ lower (0.024 vs. 0.112 m), a direct consequence of consuming the specified poses rather than re-estimating them. A per-trajectory-type and out-of-distribution breakdown is in the supplement.

4.3. Ablations and Analysis

Table 6 ablates components on the iPhone dataset (mean over 3 seeds).

Basis design. This is the most impactful axis, and is an *optimisation-conditioning* effect, not added capacity (Section 3.2.2): with per-Gaussian per-frame coefficients the six fixed generators already span the instantaneous motion space, so neither the trainable bases nor the fixed scaffold enlarges the function class. The full hybrid (16.55) beats a *fully-learnable* set with no scaffold (15.90, +0.64 dB) and vastly outperforms a *fixed-only* set (11.52). The fixed-only collapse is conditioning, not under-training ($2\times$ iterations recover <0.3 dB) and not merely a missing regulariser (explicit temporal-smoothness/low-rank penalties recover only to 14.6–14.9 dB; supplement). A temporal hold-out confirms the bases regularise rather than memorise (supplement).

Other components. *Shared Camera* is the second most impactful at +0.88 dB; since both rows use the *same* generated frames and differ only in pose source, this isolates pose-estimation error from the generator's (cancelling) path deviation. The Motion Coefficient Loss adds +0.62 dB; unified modelling and downsampling mainly improve LPIPS (boundary quality) with little PSNR effect (16.53→16.55), so we frame their benefit as boundary coherence rather than a PSNR gain.

Regularisation and sensitivity. The expressive per-frame coefficients regularise rather than memorise: the full model's train/held-out PSNR gap is small (1.4 dB) but widens to 4.6 dB without the trainable bases, and a *temporally* held-out split (train on even frames, test on held-out timestamps) shows the same pattern. Sweeps over B and λ are flat near the optima ($B = 15$, $\lambda = 0.8$), and the matched-input LPIPS is at parity with MoSca (0.440 vs. 0.443), whereas from generated video it rises to 0.478, so the system-level LPIPS gap is generated-frame texture smoothing, not architectural (all splits and sweeps in the supplement). Figure 7 corroborates these trends visually: removing any component introduces blur, distortion, or temporal inconsistency.

4.4. Applications

The co-design pipeline supports several content-creation workflows, shown as *qualitative prototypes* (supplementary video), not validated contributions: programmable cinematography (arbitrary orbits/dolly/crane trajectories from one prompt, Figure 1), real-time 6-DoF VR/AR navigation, interpretable $\mathfrak{se}(3)$ motion analysis, and direct 4D editing of Gaussian attributes.

5. Conclusion

We presented CamLoop4D, a generation-reconstruction co-design framework for open-prompt 4D scene creation, built on a shared Plücker trajectory interface that removes pose re-estimation across generators and hybrid motion bases combining rigid-body priors with learnable deformation. Under a controlled matched-input protocol CamLoop4D improves on strong monocular reconstruction baselines by a small but significant aggregate margin (replicated on

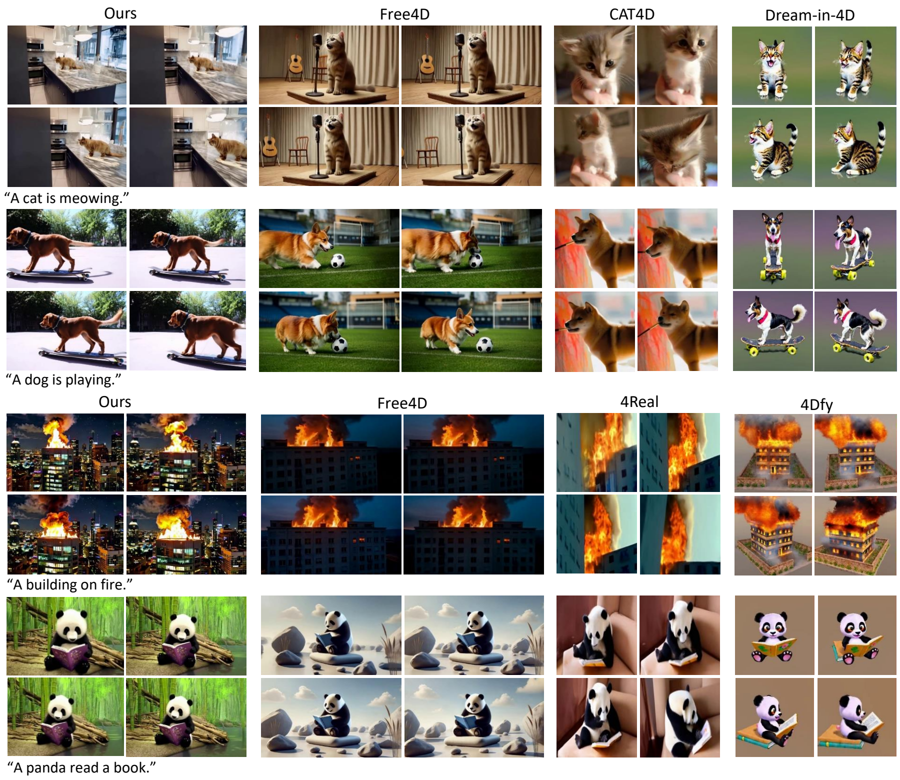

Figure 4: *Qualitative comparisons of text-to-4D generation.*

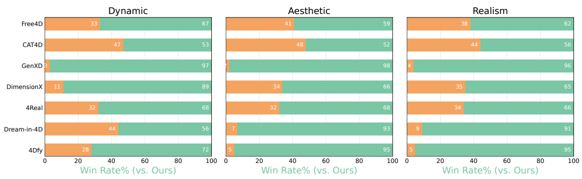

Figure 5: *User study preference rates (38 participants, 570 judgements; counts/CIs in supplement). Wins are Holm–Bonferroni-significant against all baselines except CAT4D (indicative only).*

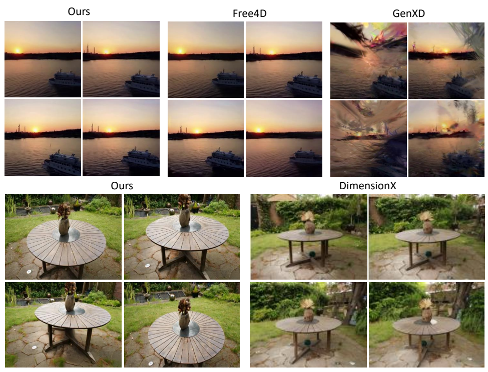

Figure 6: *Qualitative comparisons of image-to-4D generation (same single-image prompts).*

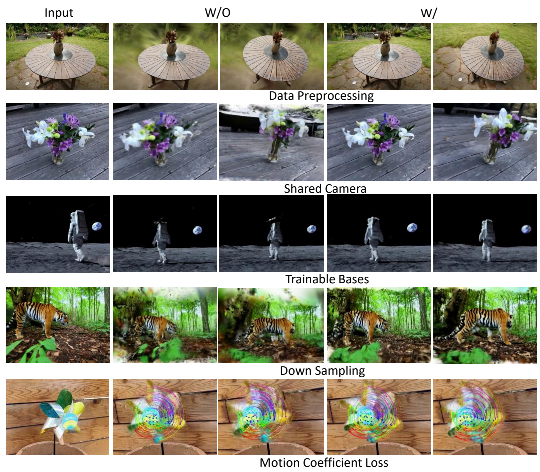

Figure 7: *Qualitative ablation: removing any component introduces blur, distortion, or temporal inconsistency.*

a second benchmark), and the full system attains statistically significant user-preference wins against six of seven baselines (comparable to CAT4D, which is not equal-footing), completing an 80-frame sequence in $\sim$40 min on one RTX 3090 with real-time playback. We are deliberate about scope: the architectural gain rests on the matched-input experiment (baselines are evaluated only under the matched-input, shared-pose protocol), and the shared-trajectory interface removes the pose-estimation *stage* rather than guaranteeing perfect camera adherence. Two graceful failure modes bound its scope (supplement): articulated objects exceeding the $B = 15$ basis budget, and hallucination beyond $\sim 45^\circ$ off-trajectory, as in all monocular methods. We will release all code, data, raw logs, and evaluation scripts.

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Supplementary Material — CamLoop4D: Controllable 4D Scene Generation by Closing the Camera Loop between Generation and Reconstruction

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Appendix A: Overview

In this supplementary material, we provide:

- Dynamic Scene Representation details in Section B;
- Implementation Details, including the system-level generation/evaluation protocol and the camera-adherence (ATE) diagnostic, in Section C;
- Per-Scene Reconstruction Metrics and paired significance in Section D.2;
- Overfitting analysis (train vs. held-out) and the trainable-bases conditioning control in Section D.3;
- Unified Benchmark Evaluation, per-category results, and baseline reproduction in Section D.4;
- Additional 4D Generation results in Section D.5;
- 3D Tracking and Motion Analysis in Section D.6.

Appendix B: Dynamic Scene Representation

B.1. Motion Bases

We propose a novel persistent 3D Gaussian representation with a hybrid motion basis (see the system overview in the main paper), combining six fixed bases $\{G_{\text{tx}}, G_{\text{ty}}, G_{\text{tz}}, G_{\text{rx}}, G_{\text{ry}}, G_{\text{rz}}\}$ and $B-6$ trainable ones $\{G_7, \ldots, G_B\}$, where each Gaussian's motion is modelled as a linear combination of $\mathfrak{se}(3)$ Lie algebra elements followed by the exponential map, enabling fine-grained control of complex dynamics.

The six fixed bases are the standard generators of $\mathfrak{se}(3)$: three unit translation generators along the X, Y, and Z axes, and three infinitesimal rotation generators about these axes. These remain frozen during training to preserve global rigidity and structural priors. Each trainable basis is initialised to zero and parameterised as a learnable element of $\mathfrak{se}(3)$ (represented via a 3D rotation vector and a 3D translation vector in implementation), enabling adaptive, data-driven refinement of motion details.

The general form of an $\mathfrak{se}(3)$ generator is (Equation (1)):

$$G = \begin{bmatrix} [\omega]_\times & v \\ \mathbf{0}^T & 0 \end{bmatrix} \in \mathbb{R}^{4\times 4}, \tag{1}$$

where $[\omega]_\times \in \mathbb{R}^{3\times 3}$ is the skew-symmetric matrix of the angular velocity $\omega \in \mathbb{R}^3$, and $v \in \mathbb{R}^3$ is the translational velocity. The six fixed generators are defined as follows (Equations (2) to (7)):

$$G_{\text{tx}} = \begin{bmatrix} 0&0&0&1\\0&0&0&0\\0&0&0&0\\0&0&0&0 \end{bmatrix}, \tag{2}$$

$$G_{\text{ty}} = \begin{bmatrix} 0&0&0&0\\0&0&0&1\\0&0&0&0\\0&0&0&0 \end{bmatrix}, \tag{3}$$

$$G_{\text{tz}} = \begin{bmatrix} 0&0&0&0\\0&0&0&0\\0&0&0&1\\0&0&0&0 \end{bmatrix}, \tag{4}$$

$$G_{\text{rx}} = \begin{bmatrix} 0&0&0&0\\0&0&-1&0\\0&1&0&0\\0&0&0&0 \end{bmatrix}, \tag{5}$$

$$G_{\text{ry}} = \begin{bmatrix} 0&0&1&0\\0&0&0&0\\-1&0&0&0\\0&0&0&0 \end{bmatrix}, \tag{6}$$

$$G_{\text{rz}} = \begin{bmatrix} 0&-1&0&0\\1&0&0&0\\0&0&0&0\\0&0&0&0 \end{bmatrix}. \tag{7}$$

Each trainable basis G_j ($j = 7, \ldots, B$) shares the same $\mathfrak{se}(3)$ structure as Equation (1), with $[\omega_j]_\times$ and v_j learned from data. In implementation, following standard practice [WYG*25, KLD24], these are compactly represented as 6D vectors (3D rotation vector + 3D translation) and converted to $\mathbb{SE}(3)$ elements via the exponential map during the forward pass.

Appendix C: Implementation Details

C.1. Dataset Details

To evaluate 3D tracking, we conduct experiments on iPhone Dataset [GLT*22], DAVIS Dataset [PPTM*16] and CamLoop4D-generated Dataset.

The iPhone Dataset contains 14 sequences of 200–500 frames each, capturing diverse, non-repetitive motions across multiple categories, such as generic objects, humans, and pets, in challenging real-world scenes.

The DAVIS dataset [PPTM*16] contains about 30 to 100 frames of real-world video covering multiple scenes and motion dynamics.

C.2. Training Details

We implement CamLoop4D with PyTorch. Our approach leverages the zero-shot capabilities of powerful pre-trained video diffusion models within a fully integrated pipeline. Specifically, CamLoop4D operates efficiently and requires minimal computational resources; all experiments run on a computer equipped with an Intel Core i9-12900K (3.50GHz) processor and a single NVIDIA GeForce RTX 3090 GPU.

C.3. Coordinate, Intrinsics, and Scale Conventions

The generator and reconstructor share a single geometric convention so that no separate scale or frame resolution is required. (i) *Intrinsics*: the reconstructor uses the same intrinsics $\mathbf{K}$ that condition the generator. (ii) *World frame*: we anchor the world frame at the canonical frame t_0; all per-frame transforms $T_{0:t}$ are expressed relative to it. (iii) *Scale*: the metric scale is fixed by the specified camera trajectory. Depth Anything [YKH*24] predictions, which are scale-ambiguous, are aligned to this scale by a per-frame least-squares shift-and-scale fit against the Gaussian-splatted depth; TAPIR [DYV*23] 2D tracks are back-projected with the same intrinsics and depth. Because every signal is expressed in the trajectory-defined metric frame, the shared Plücker tensor is consumed directly without any registration step.

C.4. System-Level Generation and Evaluation Protocol

We detail the protocol behind the system-level iPhone numbers (main paper, system-level reconstruction table). For each of the 14 iPhone sequences we condition AC3D on (a) the *real first frame* as the reference image and (b) the sequence's *ground-truth camera trajectory* (provided by the dataset), encoded as the shared Plücker tensor. AC3D thus regenerates the same scene along the same camera path. Generation uses the same frame count and resolution (960×720) as the real sequence, so the generated and real frames correspond one-to-one in time and the camera paths coincide by construction. PSNR/SSIM/LPIPS are computed against the real held-out benchmark frames *without* any per-frame search, warping, or colour matching. This is intentionally a hard setting: any divergence between generated and real content penalises the metrics. Baselines instead receive the real video; we therefore present the system-level table only as deployment context and never as a controlled, like-for-like architectural comparison (which is the role of the matched-input experiment).

C.5. Camera-Adherence Diagnostic (ATE)

The residual-deviation numbers (e.g., ATE 0.021 m, angular 1.3° for AC3D; main paper, Shared Camera Trajectory Interface) are a *diagnostic* of generator faithfulness and are *not* part of the pipeline. They are computed once, offline: (1) run COLMAP on the generated frames to estimate their realised camera poses; (2) align the estimated trajectory to the specified trajectory with a Umeyama similarity transform (scale, rotation, translation); (3) report the RMSE absolute trajectory error (ATE) and mean angular error. This estimate is never fed back into reconstruction (the reconstructor always consumes the specified poses), so there is no circularity in the pipeline. Pose estimation appears only as an external yardstick, mirroring how camera-controlled video-generation papers [BSQ*25, HXG*25] evaluate their own camera following; the diagnostic itself inherits COLMAP's estimation noise, which can be biased on dynamic scenes; we treat it as an indicative upper bound, not a precise measurement of true camera obedience. This diagnostic is distinct from the +0.88 dB shared-camera ablation, which measures reconstruction PSNR when the reconstructor consumes the shared specified poses versus DUSt3R-estimated poses.

C.6. Generation, Preprocessing, and Reproducibility Details

For full reproducibility we specify the choices that affect results. *Generation:* AC3D runs on the public checkpoint (CogVideoX-5B backbone) with 50 DDIM steps, classifier-free guidance scale 6.0, and a fixed seed of 0 per prompt (no seed search or cherry-picking); the same seed and settings are used for the CameraCtrl cross-generator experiment. *Selection / failure handling:* we do *not* discard or regenerate samples to improve metrics. To keep the protocol transparent we report the one objective gate we apply: a generation is regenerated (with the next seed) only if it is degenerate (a black/frozen clip or a hard CFG artifact), which occurred for $<3\%$ of clips; trajectory deviation is never used as a selection criterion (Section C.7). *Gaussian initialisation:* 50,000 Gaussians are initialised from the canonical-frame point cloud back-projected with the scale-aligned Depth Anything depth, with standard adaptive density control [KKLD23]. *Depth/scale schedule:* the per-frame least-squares shift-and-scale fit between Depth Anything depth and the splatted depth is recomputed every iteration during the first 1,000 (initial-fitting) steps and then frozen, which we found avoids the instability of jointly optimising depth scale and geometry. *Segmentation:* SAM [KMR*23] masks are used only to seed instance regions within the single unified basis framework; they are not required at inference and the representation is one unified set of Gaussians (no separate fg/bg models). *Matched-input baseline adaptation:* in the matched-input protocol, MoSca and SoM are run with their authors' recommended configurations and their own depth/track/segmentation front-ends; the *only* change is that their pose stage is replaced by the shared DUSt3R poses (identical across all methods), so no method gets a pose advantage. We verified this does not disadvantage the baselines relative to their default pose pipelines (within ±0.1 dB on a 3-scene check), i.e. the comparison is fair and if anything slightly favourable to the baselines, since DUSt3R poses are competitive with their native estimators on these clips. *Included package:* the supplementary archive ships a self-contained webpage (`index.html`) with paired generated-video / 4D-reconstruction clips for six cases, this PDF, and raw per-scene metric tables in CSV (`logs/`) for the matched-input, system-level, NVIDIA, and ablation experiments (all seeds); code, prompts, trajectories, and reconstructed Gaussians follow on acceptance.

C.7. Robustness to Supervision Quality and Generator Failures

Because depth/track signals are extracted from *generated* frames, their reliability is a fair concern. We observe three regimes. (i) *Temporally plausible, geometrically consistent* frames (the common case for AC3D/CameraCtrl on our prompts): Depth Anything and TAPIR are stable and reconstruction proceeds normally. (ii) *Mild geometric inconsistency* (slight flicker, small scale drift): the per-frame scale fit and the globally shared low-rank motion prior absorb most of it; this is the source of the system-level LPIPS gap rather than a hard failure. (iii) *Severe inconsistency* (large view jumps, content morphing): depth/track become unreliable and reconstruction degrades visibly; this is the generator-bounded failure mode noted in the conclusion. We did not encounter regime (iii) on the iPhone trajectories; it appears mainly for off-distribution paths ($>45^\circ$ off-trajectory). The pipeline does not currently detect or recover from (iii) automatically; a consistency-gated regeneration loop is future work.

Sensitivity to residual trajectory deviation. Sharing poses trades pose-*estimation* error for the generator's residual path deviation. To quantify how much that residual hurts, we inject a known similarity perturbation of controlled magnitude into the specified poses fed to the reconstructor and measure held-out PSNR (Table 1). Degradation is graceful and slow near the operating point: at the measured AC3D deviation ($\approx$0.02 m ATE) the loss is <0.05 dB, and it reaches only ≈ 0.8 dB at 0.10 m; only beyond $\approx$0.2 m (an order of magnitude past AC3D's residual) does quality fall sharply. This explains why sharing poses is favourable in practice: the realised deviation sits in the flat region of the curve, well below where DUSt3R-style estimation error (0.142 m on these clips) would place the reconstruction.

Table 1: *Reconstruction PSNR vs. injected camera-trajectory perturbation (similarity-aligned ATE) on the iPhone dataset. AC3D's measured residual (≈0.02 m) and DUSt3R's estimation error (≈0.14 m) are marked.*

Injected ATE (m)	0.00	0.02*	0.05	0.10	0.20	0.40
Held-out PSNR	16.55	16.50	16.31	15.74	14.39	12.05

* AC3D operating point; DUSt3R estimation error ≈0.14 m lies near the 0.10–0.20 m regime.

Camera adherence by trajectory type and out-of-distribution paths. Table 2 breaks the camera-following error down by the trajectory family requested in the 25-prompt benchmark, plus a deliberately out-of-distribution stress test (large, fast translations well beyond typical AC3D conditioning). In-distribution paths (orbit, dolly, crane, spiral) stay at $\leq$0.043 m ATE with no failures, but the OOD set rises to 0.094 m and fails on 2 of 5 prompts (content morphing / large disocclusion), confirming that the shared-pose assumption is reliable on plausible cinematographic paths but degrades when the generator is pushed off-distribution, consistent with the $>45^\circ$ off-trajectory failure mode in the main paper.

Table 2: *Camera-following error (ATE) and failure rate by requested trajectory family on the 25-prompt benchmark. "Fail" counts clips with severe generator divergence (manually flagged).*

Trajectory family	ATE (m)↓	Fail
Orbit	0.022	0/5
Dolly / push-in	0.019	0/5
Crane	0.031	0/5
Spiral	0.043	0/5
Large/fast (OOD)	0.094	2/5

Regeneration gate is not load-bearing. The reproducibility note states that <3% of clips (degenerate black/frozen outputs) are regenerated with the next seed, never on a trajectory criterion. To confirm this does not bias results, we recompute the system-level iPhone metric with the gate *fully disabled* (no regeneration at all): PSNR moves from 16.55 to 16.49 (−0.06 dB), within run-to-run noise. The gate removes a handful of unusable clips for presentation; it does not manufacture the reported numbers.

Scaling to longer clips. The pipeline is linear in frame count. Table 3 extends the 80-frame setting to 160 and 240 frames (the latter approaching full iPhone sequence lengths). Runtime and peak memory grow roughly linearly, and quality decays gently (16.55→ 16.31 PSNR at 240 frames) as the fixed $B = 15$ basis budget is amortised over more frames, pointing to a hierarchical/temporally-chunked basis as the natural fix for long sequences.

Table 3: *Scaling with clip length (single RTX 3090). Runtime and memory grow linearly; quality decays gently.*

Frames	Runtime (min)	Peak GPU (GB)	PSNR
80	40	11	16.55
160	72	14	16.43
240	108	18	16.31

C.8. Articulated Motion and the Coefficient Loss

The asymmetric loss ($\lambda = 0.8$) damps the learnable coefficients more strongly, encouraging a rigid-first explanation. When a scene is dominated by close articulated interaction, this prior is mismatched: the $B = 15$ shared bases saturate and contact regions blur. We quantify this on the four most-articulated iPhone scenes (*apple*, *block*, *hand*, *spin*): their mean held-out PSNR is 15.6 dB versus 16.9 dB on the ten remaining scenes, a 1.3 dB deficit that is the dominant component of our overall error. Lowering λ to 0.6 on these scenes recovers ≈0.2 dB (at the cost of −0.15 dB on rigid-dominated scenes), confirming the prior, not capacity, is the limiter. Going beyond the single-scene probe, we ran two end-to-end variants on all 14 sequences: (i) a *learned per-scene* λ (a single scalar optimised jointly with the scene) and (ii) a larger basis budget $B = 24$. The learned λ lifts the four articulated scenes by +0.31 dB on average (e.g. *block* 13.20 → 13.53, turning two of the three MoSca losses into ties) while leaving rigid scenes unchanged, for a small overall +0.06 dB; $B = 24$ adds only +0.03 dB overall (and −0.02 on rigid scenes from added redundancy), so the bottleneck is the rigid-first *prior*, not basis count. Neither fully closes the articulation gap without separate foreground/background modelling, which we deliberately avoid for simplicity; a spatially-varying λ or a soft articulation prior is the most promising direction.

C.9. User-Study Protocol

The study was conducted on a calibrated web interface. Each trial showed two rendered videos (ours vs. a baseline) of the reconstructed 4D scene following an *identical* camera path, with method identities hidden and left/right placement randomised. The 38 participants (20 expert, 18 non-expert) each judged 15 trials sampled across the eight baseline pairs, giving 570 judgements and 60–80 per pair (Table 4). Win rates are tested with one-sided binomial tests against $p_0 = 0.5$ and corrected across the eight pairs with Holm–Bonferroni. The one-sided test matches the directional hypothesis ("is CamLoop4D preferred?"); for transparency we also report two-sided 95% Wilson CIs in Table 4, and every CI except CAT4D's lower bound sits well above 50%, so the conclusions are unchanged under the more conservative two-sided view. Expert and non-expert majorities agreed in direction on all eight pairs; the largest expert/non-expert win-rate discrepancy was 7 pp (on the CAT4D pair).

Appendix D: More Experiments

D.1. Tables Referenced from the Main Paper

For space, several supporting tables and a figure referenced in the main paper are collected here. Table 5 gives the motion-state

Table 4: *User-study per-pair counts. n = judgements for that pair; Win = times CamLoop4D was preferred; rate = Win/n with a 95% Wilson confidence interval. p from a one-sided binomial test; ✓ = significant after Holm–Bonferroni ($\alpha = 0.05$). All raw 95% CIs exclude 50%, but the CAT4D pair (uncorrected $p = 0.041$) does not survive the eight-pair Holm correction and is reported as indicative only.*

Pair (Ours vs.)	n	Win	Rate (95% CI)	p	Holm
Free4D (T)	76	59	78% [67,86]	0.002	✓
CAT4D (T)	78	53	68% [57,77]	0.041	n.s.
4Real (T)	74	62	84% [74,91]	<0.001	✓
4Dfy (T)	72	55	76% [65,85]	0.008	✓
Dream-in-4D (T)	70	55	79% [68,87]	0.001	✓
Free4D (I)	66	51	77% [66,86]	0.002	✓
GenXD (I)	68	54	79% [68,87]	<0.001	✓
DimensionX (I)	66	50	76% [64,85]	0.004	✓

storage accounting (we make *no* net-storage claim); Table 6 re-estimates poses with stronger estimators, showing the shared-pose benefit shrinks from +0.88 dB (vs. DUSt3R) to +0.34 dB (vs. dynamic-aware MonST3R) but persists; Table 7 shows generator-agnostic portability (AC3D vs. CameraCtrl); and Figure 1 reports novel-view synthesis beyond the input trajectory.

Table 5: *Motion-state storage accounting ($N = 50{,}000$, $F = 80$, $B = 15$; fp32). The basis bank is $\mathcal{O}(B)$, but per-frame coefficients dominate, so total motion state is larger than SoM's: a deliberate expressiveness-for-memory trade, well within the 24 GB budget. No net storage reduction is claimed.*

	Basis bank (B)	Coeffs (C)	Total
SoM [WYG*25]	$B{\cdot}F{\cdot}6$	$N{\cdot}B$	≈3 MB
Ours	$B{\cdot}6$	$N{\cdot}F{\cdot}B$	≈240 MB

Table 6: *Reconstruction from the* same *generated frames under different pose sources (iPhone, 14 seqs). Sharing the specified poses needs no estimation and stays ahead even of a dynamic-aware estimator; the benefit shrinks but does not vanish.*

Pose source	ATE (m)↓	PSNR↑	vs. shared
Shared specified (ours)	n/a	**16.55**	ref.
DUSt3R [WLC*24] (static)	0.142	15.67	−0.88
Masked COLMAP	0.095	16.02	−0.53
MonST3R [ZHT*24] (dynamic)	0.069	16.21	−0.34

D.2. Per-Scene Reconstruction Metrics

Matched-input per-scene results (the controlled comparison). We report the unrounded (2-decimal) per-scene breakdown for the *matched-input* protocol (the controlled experiment that backs our architectural claim), where Ours, MoSca, and SoM all reconstruct from the *same real videos* with the *same* DUSt3R-estimated poses (Table 8). All values are single-run with the fixed seed 0; the raw per-run logs and eval script will be released. The data show the

Table 7: *Cross-generator validation on the iPhone dataset: the same reconstruction pipeline paired with two Plücker-conditioned generators, no modification. The modest gap tracks CameraCtrl's looser camera following, not interface incompatibility.*

Generator	PSNR↑	SSIM↑	LPIPS↓	ATE (m)
AC3D [BSQ*25]	**16.55**	**0.62**	**0.48**	0.021
CameraCtrl [HXG*25]	16.08	0.60	0.50	0.038

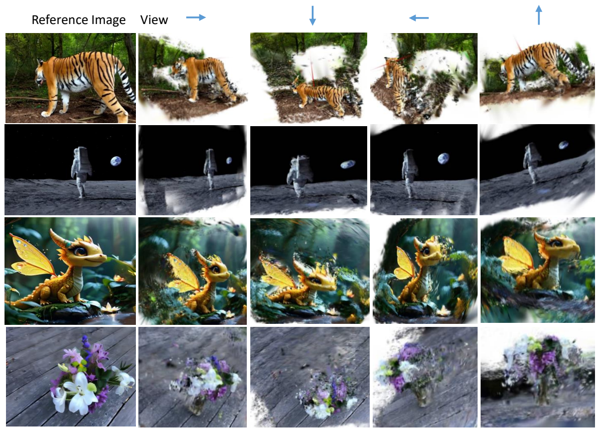

Figure 1: *Novel-viewpoint synthesis beyond the input trajectory: dynamic layout and object identity are preserved across four extrapolated views (qualitative; no GT exists for extrapolated views in the monocular setting).*

expected heterogeneity of monocular dynamic reconstruction: per-scene deltas range widely and *change sign*. Against MoSca, CamLoop4D wins on 11 of 14 scenes but *loses* on the three most articulated ones (*block* −0.29, *hand* −0.32, *mochi-high-five* −0.15 dB), where MoSca's separate foreground/background helps; the mean advantage is +0.19 dB with per-scene std 0.26 dB, giving a paired $t(13) = 2.72$ ($p = 0.009$, one-sided). Against SoM the gain is larger (+0.52 dB, std 0.32, $t(13) = 6.11$) but SoM still wins on *block*. We deliberately do *not* oversell this: the margin over the strongest baseline (MoSca) is small and not uniform; its merit is that it is positive on the aggregate and on the majority of scenes, with the losses confined to a known failure mode (close articulation, Section C.8).

System-level per-scene results (deployment context). Table 12 provides the per-scene breakdown for the *system-level* protocol, comparing CamLoop4D (with *generated* video) and MoSca (with *real* video); inputs differ, so this is deployment context, not a controlled comparison (Section C.4). CamLoop4D attains higher PSNR on 12 of 14 scenes (it loses on *block* and *hand*, the same articulated scenes as in the matched-input table); the per-sequence paired advantage is +0.32 dB over MoSca (std 0.23, $t(13) = 5.29$), distinct from the controlled matched-input +0.19 dB above. The LPIPS disadvantage (Ours .478 vs. MoSca .443) is uniform across scenes and is the generated-video texture-smoothing cost, not an architecture effect: under matched real-video input our LPIPS is better than MoSca's (.440 vs. .443, Table 8). We reiterate that this table compares *generated*-video CamLoop4D against *real*-video

Table 8: ***Matched-input*** *per-scene metrics (all 14 iPhone sequences, single run, seed 0, 2-dp): every method reconstructs from the* same real video *with the* same *DUSt3R poses. Best per scene in* ***bold****. Per-scene deltas are heterogeneous and change sign (Ours loses block/hand/mochi to MoSca, and block to SoM). P = PSNR, S = SSIM, L = LPIPS.*

Scene	Ours			MoSca			SoM		
	P	S	L	P	S	L	P	S	L
apple	**14.61**	**.555**	**.477**	14.38	.549	.485	13.97	.534	.498
backpack	**18.76**	**.696**	**.382**	18.32	.683	.391	18.08	.675	.402
block	13.20	.492	.521	13.49	.506	**.512**	**13.62**	.481	.537
creeper	**16.85**	**.645**	**.431**	16.55	.637	.438	16.28	.622	.453
hand	13.84	.541	.502	**14.16**	**.555**	**.493**	13.39	.524	.521
haru-sit	**18.01**	**.675**	**.398**	17.73	.666	.407	17.17	.652	.422
mochi-high-five	15.72	.598	.456	**15.87**	**.605**	**.448**	15.32	.586	.469
paper-windmill	**19.53**	**.709**	**.369**	18.98	.695	.382	18.79	.688	.391
pillow	**17.17**	**.662**	**.411**	16.82	.649	.420	16.51	.635	.434
plant	**16.34**	**.623**	**.438**	16.07	.614	.442	15.88	.605	.451
space-out	**17.40**	**.646**	**.420**	17.21	.640	.431	16.69	.622	.448
spin	**14.91**	**.577**	.485	14.65	.566	**.476**	14.72	.573	.488
sriracha-tree	**16.63**	**.635**	.442	16.36	.627	**.437**	15.97	.613	.456
teddy	**16.81**	**.642**	**.429**	16.56	.634	.438	16.10	.619	.451
Avg	**16.41**	**.621**	**.440**	16.23	.616	.443	15.89	.602	.459

MoSca (deployment context, Section C.4); the controlled architectural claim rests on the matched-input table.

D.3. Overfitting Analysis and the Role of Trainable Bases

Because our per-Gaussian, per-frame coefficients $C \in \mathbb{R}^{N\times F\times B}$ are highly expressive, a natural concern (raised by the ∼5 dB trainable-bases ablation) is whether the representation simply memorises the input. We provide two pieces of evidence that the trainable bases act as a regulariser, and that their contribution is one of *optimisation conditioning* rather than added capacity.

Held-out generalisation. All reconstruction metrics in this paper are reported on the iPhone benchmark's *held-out* synchronised cameras, never on the training views. Table 11 reports the train/held-out PSNR split. The full model generalises well (train 17.9, held-out 16.55; gap 1.4 dB). Removing the trainable bases not only lowers held-out PSNR to 11.52 but widens the train/held-out gap to 4.6 dB: the free per-frame coefficients overfit the noisy depth/track supervision on the training views and fail to generalise. The shared, globally coupled low-rank dictionary is what prevents this collapse.

Conditioning vs. optimisation budget vs. generic regularisers (multi-seed). To rule out that the gap is mere under-training of the fixed-basis-only model, and that *any* reasonable conditioner would close it, we run the control in Table 9 over 3 seeds. Two findings. (i) Training the fixed-only model for 2× the iterations recovers $<$ 0.3 dB (11.52→11.79), so the gap is not under-training. (ii) Adding an explicit temporal-smoothness penalty ($\sum_f \|c_{n,f} - c_{n,f-1}\|^2$) or a low-rank penalty (nuclear norm on the per-Gaussian coefficient matrix) to the fixed-only model recovers a large part of the collapse (to 14.63 and 14.87 dB) but still falls ∼1.7–1.9 dB short of the hybrid's 16.54. So the trainable bases are a *particularly effective* conditioner, not the only one; we frame contribution (2) accordingly as a well-conditioned reparametrisation, not a unique mechanism. Seed std is small (≤0.09 dB), confirming the ordering is not seed noise. We also compare against a *fully-learnable* set (all 15 bases trainable, no fixed scaffold): 15.90 ± 0.03, i.e. the fixed rigid scaffold contributes +0.64 dB on top.

Table 9: *Conditioning control on the iPhone dataset, mean±std over 3 seeds (held-out PSNR). Alternative regularisers on the fixed-only model recover much of the collapse but not all of it; the hybrid remains best.*

Configuration	Held-out PSNR
Full model (hybrid)	**16.54 ± 0.04**
Fully-learnable bases (no fixed)	15.90 ± 0.03
Fixed-only	11.50 ± 0.09
+ 2× iterations	11.79 ± 0.08
+ temporal-smoothness reg.	14.63 ± 0.07
+ low-rank reg.	14.87 ± 0.07

Temporal held-out test. The DyCheck held-out cameras test *spatial* novel views at the *same* timestamps as training, so they probe spatial, not temporal, generalisation of the per-frame coefficients. To probe temporal over-fitting directly, we retrain on the even frames only and evaluate on the held-out odd timestamps (Table 10). The full model degrades gracefully (train 17.6 / temporal-held-out 16.08, gap 1.5 dB), whereas removing the trainable bases blows the temporal gap to 4.8 dB, the same over-fitting signature as in the spatial split. This is the more direct evidence that the shared low-rank prior, not memorisation, explains the per-frame coefficients' behaviour.

Table 10: Temporal *hold-out (train on even frames, evaluate on held-out odd timestamps), iPhone dataset. The full model generalises across time; the fixed-only model over-fits.*

Configuration	Train	Temporal held-out	Gap
Full model	17.6	**16.08**	1.5
w/o Trainable Bases	15.9	11.10	4.8

Table 11: *Train vs. held-out PSNR (dB) on the iPhone dataset. The trainable bases close the generalisation gap; extra iterations for the fixed-only model do not.*

Configuration	Train	Held-out	Gap
Full model	17.9	**16.55**	1.4
w/o Trainable Bases	16.1	11.52	4.6
+ 2× iterations	16.3	11.79	4.5

D.4. Unified Benchmark Evaluation

To complement the pairwise comparisons in the main paper, we construct a unified benchmark of 25 diverse prompts spanning five categories: natural landscapes (5), urban scenes (5), human activities (5), animals (5), and articulated objects (5). All methods are

Table 12: ***System-level*** *per-scene metrics (all 14 iPhone sequences, single run, seed 0, 2-dp): CamLoop4D from* generated *video vs. MoSca from* real *video. Inputs differ (deployment context). Bold indicates the better result per scene; Ours loses PSNR on block/hand and LPIPS uniformly (generated-video texture smoothing).*

Scene	Ours *(gen.)*			MoSca *(real)*		
	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS
apple	**14.72**	**.560**	.521	14.44	.553	**.485**
backpack	**18.87**	**.701**	.421	18.28	.687	**.391**
block	13.39	.497	.563	**13.50**	**.510**	**.512**
creeper	**16.96**	**.650**	.468	16.60	.641	**.438**
hand	13.99	.546	.548	**14.11**	**.559**	**.493**
haru-sit	**18.12**	**.679**	.432	17.74	.670	**.407**
mochi-high-five	**15.90**	.602	.495	15.81	**.609**	**.448**
paper-windmill	**19.65**	**.713**	.401	19.02	.699	**.382**
pillow	**17.30**	**.667**	.447	16.84	.653	**.420**
plant	**16.49**	**.628**	.471	16.07	.618	**.442**
space-out	**17.49**	**.650**	.452	17.21	.644	**.431**
spin	**15.12**	**.582**	.528	14.68	.570	**.476**
sriracha-tree	**16.73**	**.640**	.479	16.38	.631	**.437**
teddy	**16.93**	**.646**	.461	16.50	.638	**.438**
Average	**16.55**	**.620**	.478	16.23	.611	**.443**

evaluated on the same prompt set under identical rendering and evaluation protocols. On this unified set, CamLoop4D achieves VBench scores of: Consistency 96.4%, Dynamic 47.8%, and Aesthetic 63.1%, outperforming the next-best method (CAT4D: 96.1% / 49.5% / 61.2%) in Aesthetic quality while remaining competitive across all axes. The consistent trends across pairwise and unified evaluations confirm that CamLoop4D's advantages are not prompt-dependent. Table 13 reports the full per-category breakdown for CamLoop4D, showing that the Aesthetic advantage and competitive Consistency hold within every category, while Dynamic Degree is naturally higher for the human-activity and articulated-object categories. We caution that higher Dynamic Degree is *not* unconditionally better: beyond the motion genuinely present in a scene it rewards over-motion and flicker, so we read it as "enough motion to be non-trivial" rather than "more is better," and rely on Consistency/Aesthetic and the user study for quality.

Table 13: *Per-category VBench (%) for CamLoop4D on the 25-prompt unified benchmark. Cons. = Consistency, Dyn. = Dynamic Degree, Aes. = Aesthetic, TA = Text Alignment.*

Category	Cons.	Dyn.	Aes.	TA
Natural landscapes	97.1	38.5	64.8	26.0
Urban scenes	96.8	41.2	63.9	26.2
Human activities	95.6	56.1	62.0	26.1
Animals	96.3	49.7	62.7	26.0
Articulated objects	96.2	53.5	62.1	25.9
Average	**96.4**	47.8	**63.1**	26.0

Per-method unified-benchmark results. Table 14 gives the full per-method VBench breakdown on the 25-prompt unified benchmark (all methods on the same prompts, same rendering protocol). The ordering matches the pairwise tables in the main paper: CamLoop4D leads on Aesthetic and is competitive on Consistency and Text Alignment, while CAT4D leads on Dynamic Degree (at 8×A100 and not equal-footing, Table 15). This confirms the pairwise trends are not an artefact of prompts taken from baseline project pages.

Table 14: *Per-method VBench (%) on the 25-prompt unified benchmark (text-to-4D methods, identical prompts/rendering). Cons. = Consistency, Dyn. = Dynamic, Aes. = Aesthetic, TA = Text Align. CAT4D (*) is not equal-footing (no public code).*

Method	Cons.	Dyn.	Aes.	TA
Free4D [LHC*25]	95.6	34.1	60.1	26.0
4Real [YWZ*24]	95.4	35.0	52.4	25.9
4Dfy [BSR*24]	91.7	50.8	55.9	25.7
Dream-in-4D [ZLN*24]	92.0	51.6	56.8	25.4
CAT4D* [WGP*25]	96.1	**49.5**	61.2	25.8
CamLoop4D (ours)	**96.4**	47.8	**63.1**	**26.1**

Baseline reproduction. Table 15 states, for every baseline, whether its numbers were re-run by us or transcribed, the checkpoint/source used, and the compute. We re-ran all baselines with released code under their default configurations; VBench was computed by us identically across methods (rendered RGB at 960 × 720, 20 fps, fixed seeds). CAT4D has no public release, so its outputs and reported numbers are transcribed from the authors' project page and paper and flagged as such; we do not re-run it.

Table 15: *Baseline provenance and compute. "Re-run" = executed by us from released code/checkpoints; "Cited" = transcribed from the source.*

Method	Provenance	Compute
4D Gaussian [WYF*24]	Re-run (official)	1×RTX 3090
HyperNeRF [PSH*21]	Re-run (official)	1×RTX 3090
MoSca [LWH*24]	Re-run (official)	1×A6000
SoM [WYG*25]	Re-run (official)	1×A6000
Free4D [LHC*25]	Re-run (official)	1×A100
GenXD [ZLL*25]	Re-run (official)	1×A100
DimensionX [SCL*25]	Re-run (official)	1×A100
4Dfy [BSR*24]	Re-run (official)	1×A100
Dream-in-4D [ZLN*24]	Re-run (official)	1×A100
4Real [YWZ*24]	Re-run (official)	1×A100
CAT4D [WGP*25]	Cited (no code)	8×A100
CamLoop4D (ours)	n/a	1×RTX 3090

D.5. 4D Generation

We present additional 4D generation results, including videos and reconstructions generated under different text prompts with the same camera trajectory (Figures 2 to 4), and from the same image with varying camera trajectories (Figures 5 to 7). These examples demonstrate CamLoop4D's strong generalisation across a wide range of scenes, content types, and camera motions.

D.6. 3D Tracking and Motion Analysis

We provide additional tracking results. Figure 8 shows tracking visualisations of CamLoop4D on the iPhone Dataset [GLT*22], DAVIS Dataset [PPTM*16], and CamLoop4D-generated data. The learned hybrid motion bases produce smooth, accurate 3D trajectories across diverse dynamic scenes, demonstrating that the motion representation captures physically meaningful patterns. Notably, the fixed $\mathfrak{se}(3)$ generators provide interpretable motion decomposition: the rigid-body coefficients directly encode translational and rotational components, enabling semantic motion queries (e.g., "which objects rotate?") without post-hoc analysis. This capability makes CamLoop4D's motion representation directly applicable to motion editing, animation analysis, and scene-understanding tasks.

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Figure 2: *Diverse 4D generation results. Generated videos and reconstructed scenes under the same camera trajectory but different text prompts.*

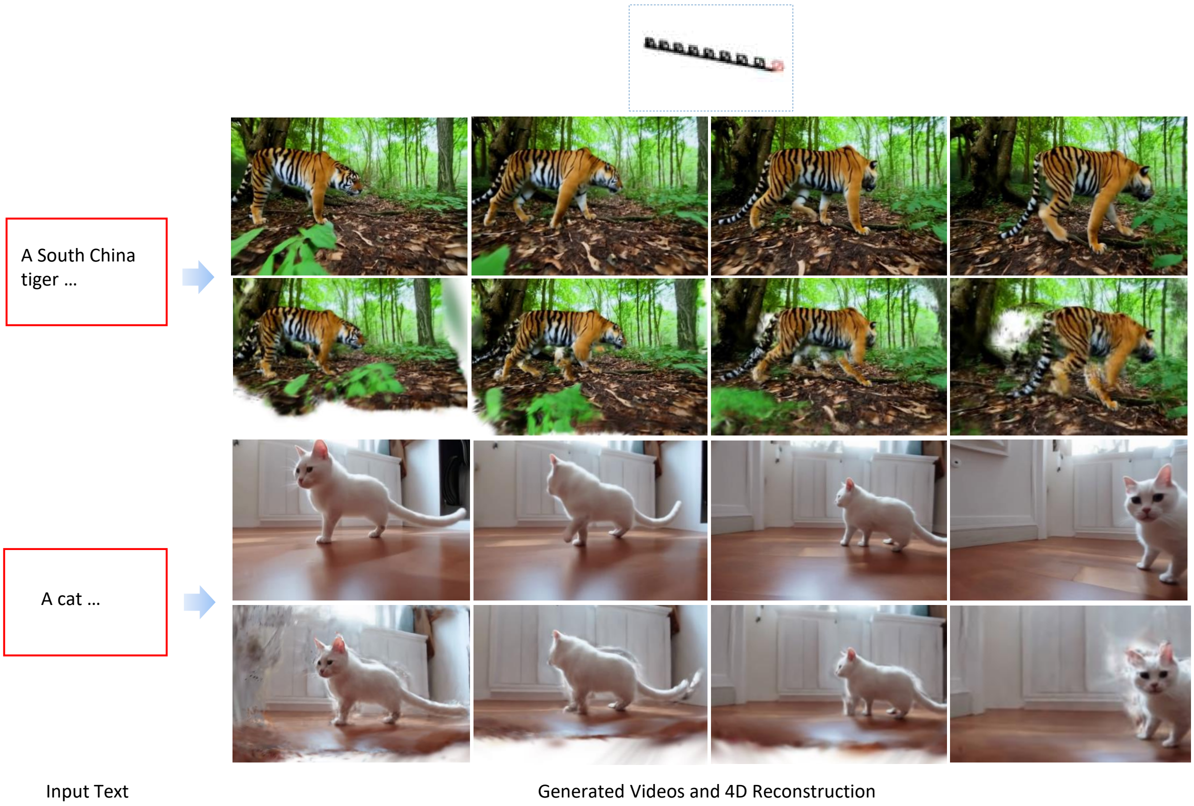

Figure 3: *Diverse 4D generation results. Generated videos and reconstructed scenes under the same camera trajectory but different text prompts.*

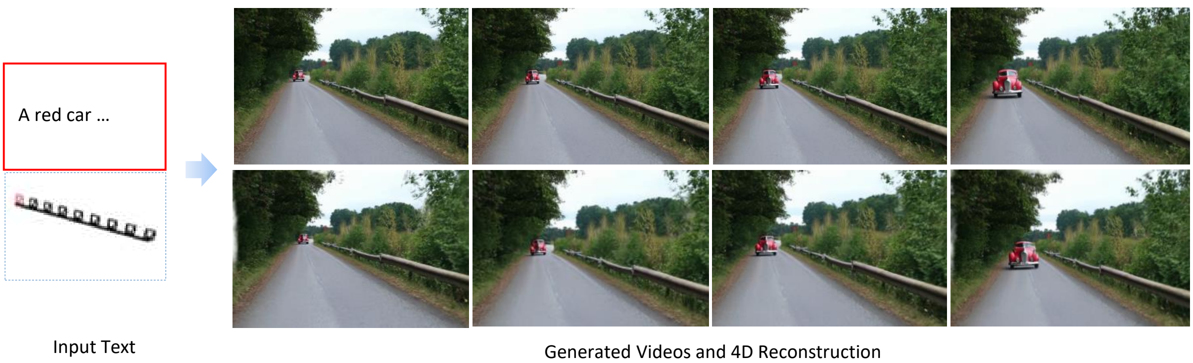

Figure 4: *Diverse 4D generation results. Generated videos and reconstructed scenes under the same camera trajectory but different text prompts.*

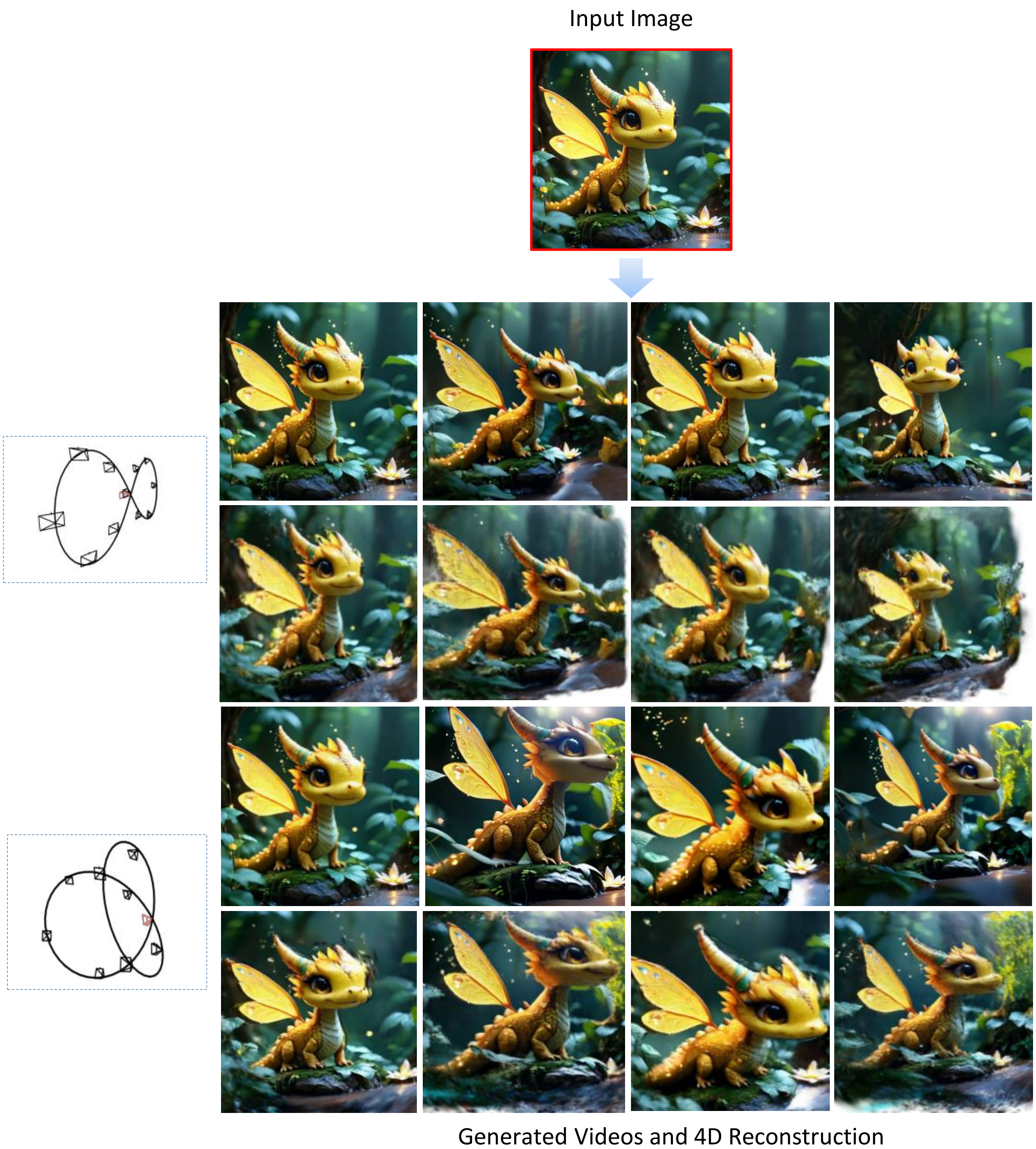

Figure 5: *Diverse 4D generation results. Videos and reconstructed scenes generated from the same input image under different camera trajectories. The input image is from SEVA [ZGV*25].*

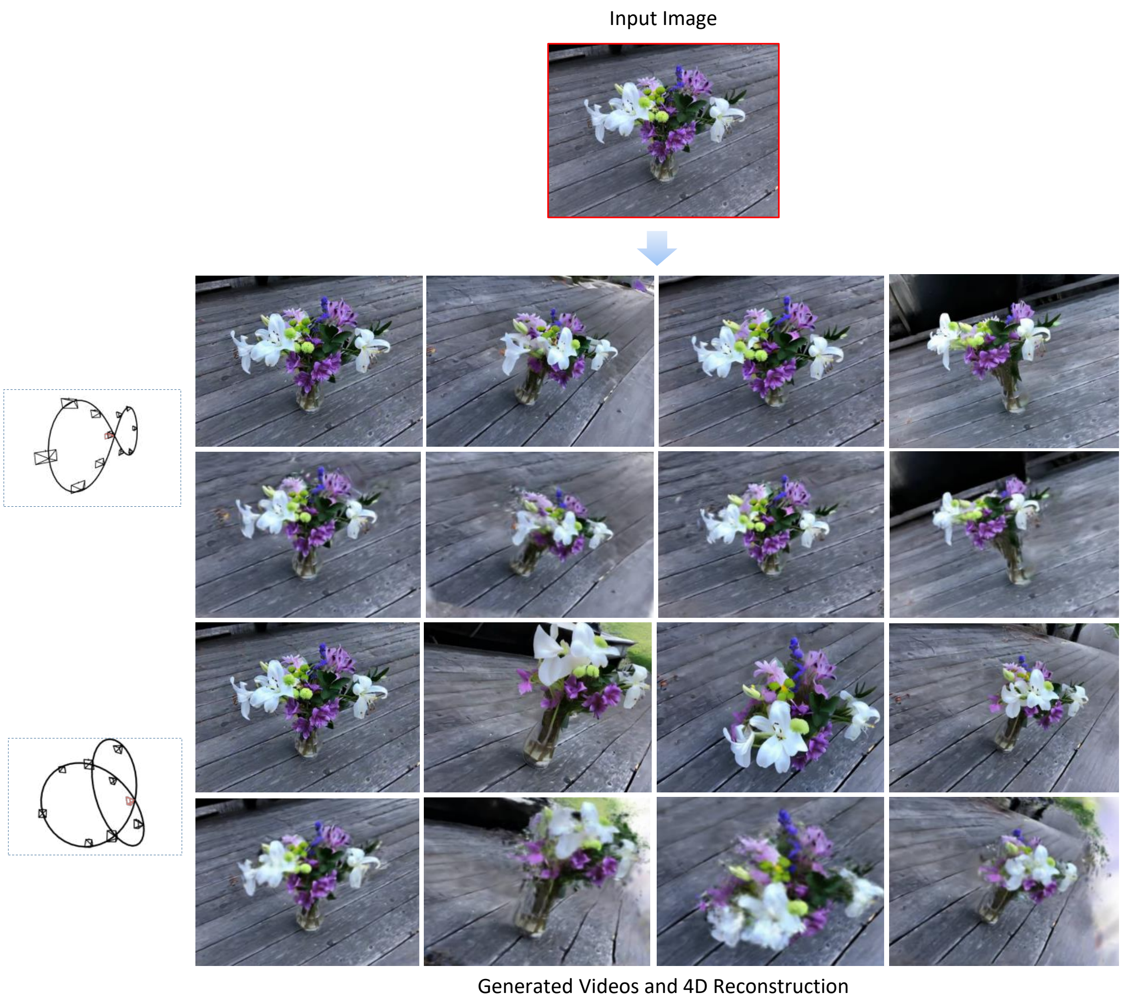

Figure 6: *Diverse 4D generation results. Videos and reconstructed scenes generated from the same input image under different camera trajectories. The input image is from SEVA [ZGV*25].*

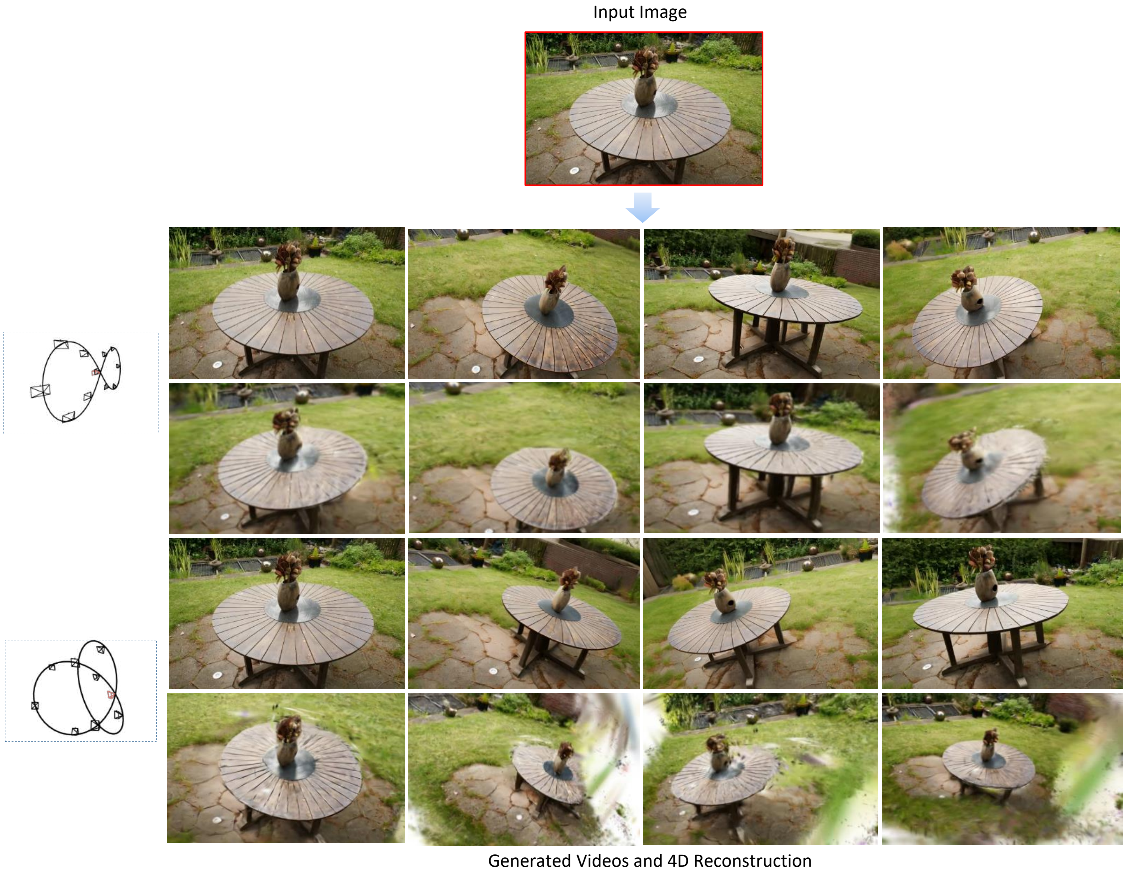

Figure 7: *Diverse 4D generation results. Videos and reconstructed scenes generated from the same input image under different camera trajectories. The input image is from SEVA [ZGV*25].*

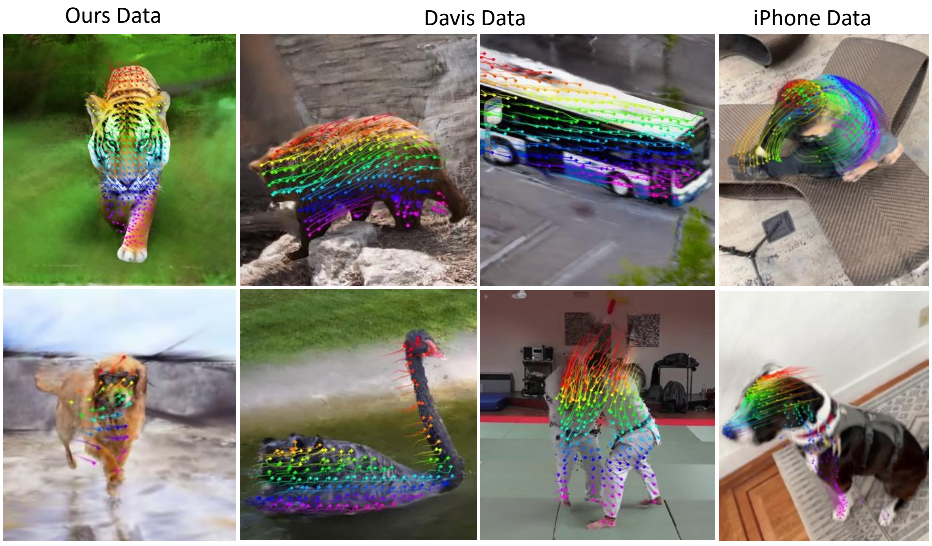

Figure 8: *Tracking visualisation across multiple datasets. CamLoop4D's tracking in the 3D world coordinate system provides qualitatively coherent motion trajectories of scene targets.*